Venkata Lakshmi Parimala Pasupuleti

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PROFESSIONAL SUMMARY

Software Engineer with 8+ years of production Python engineering and a focus on AI-powered systems. Built **AgentProbe** - an open-source multi-agent AI toolkit (LangGraph, FastAPI, ChromaDB, LLM-as-judge evaluation) live on GitHub. Shipped 20+ production modules at Entrata across 100K+ MAU - including LIHTC/Tax Credit, HUD, and Section 8/HCV affordable housing compliance automation. Built LeaseI - an agentic RAG pipeline for lease compliance. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

WORK EXPERIENCE

George Mason University

Aug 2024 -- Aug 2025

Research Associate — Fairfax, Virginia

- Built regression models ($R^2 = 0.977$) in Python and R across WHO and UK Road Safety datasets spanning 190+ countries; designed Tableau, ggplot2, and Quarto dashboards improving faculty research comprehension by 40%.
- Applied Isolation Forest to automate data-quality checks across 3 research pipelines -- surfaced anomalies manual review missed, improving downstream model reliability.

Entrata

Jan 2017 -- Aug 2024

Trainee → Associate Engineer → Software Engineer

- Built 20+ full-stack microservices (Python, PostgreSQL, FastAPI, RabbitMQ) across 120+ Agile sprints for **100K+ monthly active users** -- 40% deployment time reduction; 99.5% uptime over 6+ months; cut execution time 50% via Docker containerization.
- Pioneered Entrata's AI engineering foundation -- built and internally shipped the first AI pipeline (Python, PyTorch, OpenAI API) across **1.6M real financial records** with output validation, quality gates, and compliance controls; later productized by the team into a customer-facing feature.
- Developed **affordable housing compliance modules** (Section 8/HCV, LIHTC/Tax Credit, HUD) -- automated income eligibility verification, AMI threshold calculations, recertification workflows, and waitlist management across 100K+ unit portfolios.
- Achieved **100% PHI compliance** migrating all client data to AWS S3 with Libsodium-encrypted pipelines -- zero security incidents; improved query performance 30% across 200+ SQL queries; led 300+ peer code reviews.

PROJECTS

AgentProbe: Multi-Agent AI Testing & Evaluation Toolkit

Open Source

Python · LangChain · LangGraph · FastAPI · ChromaDB · OpenAI · Anthropic · React · Docker · GitHub Actions — github.com/venkatalakshmiparimala/AgentProbe

- Built a **3-agent LangGraph runtime** (Orchestrator, RAG Tracer, Auto Fix) with tool calling, inter-agent state handoffs, and multi-provider model routing -- deployed as a 15-endpoint FastAPI backend with CI/CD quality gates.
- Implemented **5-metric LLM-as-judge evaluation engine** (factual accuracy, hallucination, retrieval, completeness, compliance) blocking non-compliant agent runs before production; React dashboard tracking per-run metrics and regression drift.

LeaseI: AI Lease & Affordable-Housing Compliance System

Python · LangGraph · SQLite · reportlab — github.com/venkatalakshmiparimala/Lease_complaince_rag

- Built an agentic LangGraph RAG pipeline auditing residential lease PDFs for compliance gaps across **10 rules, 5 US states + federal standards** -- **93.6% precision, 82.9% recall, 87.9% F1** on 1,240 leases; deterministic rules engine keeps compliance determinations fully auditable.
- Implemented pass@k / pass^k eval harness (pass@1 81.7%, pass@3 88.3%) with multi-trial sampling and regression gate; LLM-as-judge scores both accuracy and reasoning quality on every model change.

AI on FHIR: NLP-Driven Medical Records Query System

Flask · LLMs · FHIR · React · spaCy · BERT · Embeddings · Docker — github.com/venkatalakshmiparimala/AI_on_FHIR

- Reduced clinical retrieval time ~60% building an NLP platform translating natural language queries (EN/ES) to FHIR healthcare APIs using LLMs, BERT, and semantic embeddings -- **HIPAA-compliant RBAC** across structured medical records.

CrashClarity: Large-Scale Road Accident Analysis

Python · R · Scikit-learn · Random Forest · Logistic Regression · Feature Engineering · GitHub Pages — Live Dashboard

- Achieved **85%+ accuracy** on 104K+ UK accident records by building a Random Forest + Logistic Regression pipeline with feature engineering across weather, road, and time variables -- deployed interactive dashboard on GitHub Pages.

EDUCATION

M.S. Data Analytics Engineering — George Mason University — Graduated May 2026

GPA: 3.87 · Fairfax, Virginia

B.Tech Computer Science & Engineering — Sree Venkateshwara College of Engineering — 2013 -- 2017

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL, PL/SQL, R, Bash, HTML/CSS

AI & Agents: LangChain, LangGraph, Agentic AI, Multi-Agent Systems, RAG Pipelines, Semantic Search, Embeddings, NLP, LLM Evaluation, LLM-as-Judge, Prompt Engineering, Agent Observability, ChromaDB, OpenAI API, Anthropic API, Hugging Face, FastAPI, REST APIs

ML & Data: PyTorch, TensorFlow, Scikit-learn, XGBoost, Isolation Forest, Anomaly Detection, Feature Engineering, Statistical Modeling, Model Deployment, MLOps, BERT, spaCy, NumPy, Pandas, Jupyter, Tableau, Power BI, ggplot2

Cloud & Infrastructure: AWS (S3, EC2, RDS, Glue), Azure, Docker, CI/CD (GitHub Actions), Microservices, Data Pipelines, ETL, RabbitMQ, PostgreSQL, MySQL, Oracle, Git, Linux, Agile/Scrum